Monitoring Docker with Prometheus

Prometheus is free, open source, no fuss software to monitor computer systems and alert sysadmins. It is a very vital piece of infrastructure software that informs you about the health of your systems. Prometheus can be used very effectively to monitor Docker systems.

Monitoring is an important but often overlooked feature in most applications. At the same time, it is an extremely vital part of infrastructure software – it ensures that you are aware about the health of your systems and whether the important metrics about various services running in your infrastructure are within the acceptable range.

When it comes to Docker, there have been several tools available with its toolset to monitor the health of the system, in addition to the various commercial tools in this space. Here are some expectations from a monitoring solution:

- Simple and unobtrusive to set up
- Extensive integration points and the capability to add more targets to monitor
- Open source, with a thriving community and support ecosystem
- Alerting mechanism
- Visualisation dashboards to understand the health of the system quickly

Prometheus has been gaining popularity over the last few years. It has been promoted to a top level project in the CNCF (Cloud Native Computing Foundation), and has seen significant adoption in the community, where it is used for monitoring infrastructure. It is easy to set up and has a wealth of external connectors (80+) to popular software like MySQL, Redis, NGINX and others, that make it a breeze to integrate metrics from these software into the Prometheus time series database.

Prometheus comes with its own rich metrics format, and for it to integrate any monitoring target, all the target needs to do is expose a metrics endpoint that provides it the metrics data in the Prometheus format.

Docker and Prometheus

Starting with the release of Docker 1.13, an experimental Prometheus metrics-compatible endpoint has been exposed in Docker. What this means is that Prometheus can be set up to monitor a Docker target, and it will be able to integrate the metrics endpoint into its time series database. Once the metrics have been ingested, then the